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Indian Institute of
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**Dept. of CSE - Cyber Security,
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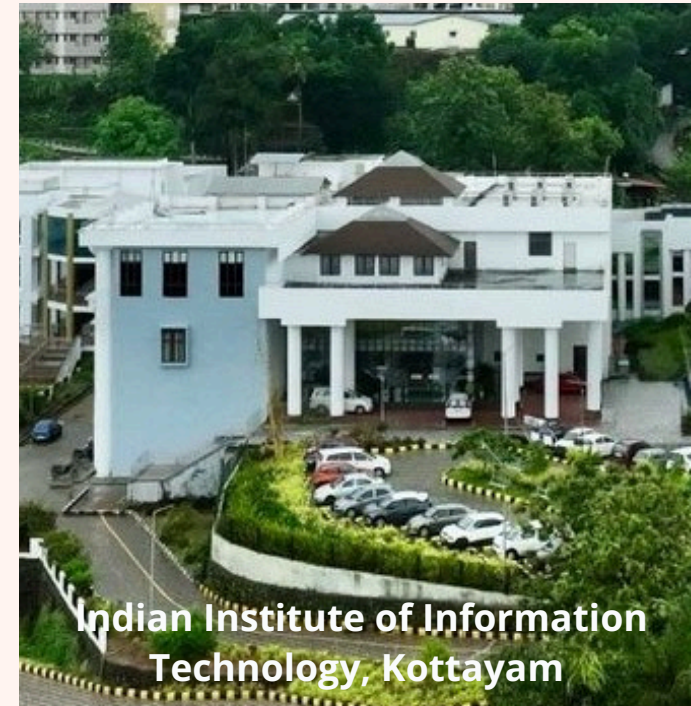
Presents

**Five Days
Hands-on FDP and
Workshop
(Online Mode)**

on

**Quantum Computing for Cyber
Security: A Hands-on Approach**

25 - 29 May 2026



**Indian Institute of Information
Technology, Kottayam**

About the Workshop

This Hands-on FDP and workshop focuses on Quantum Computing for Cyber Security, featuring hands-on sessions and relevant tools. Topics include Fundamentals of Quantum Computing covering Qubits, Superposition, Entanglement, Quantum Gates and Quantum Circuits; Quantum Key Distribution; Post-Quantum Cryptography; Quantum-resistant Algorithms; Shor's and Grover's Algorithms and their impact on classical crypto; Quantum Threat Modeling for Cloud and Distributed Systems; Quantum Random Number Generation; Secure Communication using QKD simulators; Hybrid Classical-Quantum Security Protocols; and AI-enhanced Quantum Cyber Defense.

Our skilled resource persons will walk you through the complexities of these topics, providing a thorough overview of the latest advances and research directions in quantum-safe cybersecurity.

Topics Covered

Day-1 Quantum Mechanic Basics, Qubits, Superposition, Entanglement, Quantum Gates and Quantum Circuits; Quantum Key Distribution – BB84 Protocol Hands-on; Quantum Random Number Generation using Simulators

Day-2 Shor's Algorithm and its Impact on Classical Cryptography; Grover's Algorithm for Quantum Search Attacks; Post-Quantum Cryptography – Lattice-based and Hash-based Cryptography

Day-3 Quantum Threat Modeling for Cloud and Distributed Systems; Hybrid Classical-Quantum Security Protocols; Quantum-resistant Algorithms for IoT Security

Day-4 Secure Communication using QKD Simulator Tools; AI-enhanced Quantum Cyber Defense Techniques; Quantum-safe Blockchain and Distributed Ledger Security

Day-5 Hands-on Implementation of Quantum Cryptographic Protocols; Case Studies on Quantum Cyber Attacks and Defenses; Research Directions and Future Trends in Quantum-safe Cybersecurity

Participation

Participants can join the Hands-on FDP and workshop in online mode. The workshop is open to working professionals, faculty members, research scholars in CSE/IT/Cyber Security or related fields, PG Students (MCA or M.Sc. Computer Science/IT, M.Tech/M.E. in CSE/IT/Cyber Security or related fields), and UG students (B.Tech in CSE/IT/Cyber Security or related fields) from all institutes, colleges and Universities.

UG/PG Students	1000
Research Scholars	1000
Faculty	1500
Industry	2000

Registration Fees is Non Refundable

About the Institute

The Indian Institute of Information Technology Kottayam (IIIT Kottayam) was founded in 2015 as one of the IIITs under the administrative control of the Government of India. It was later designated as an "Institution of National Importance" by an Act of Parliament enacted in 2017. The institute offers undergraduate, postgraduate and doctoral programs in Computer Science and Engineering, Electronics and Communication Engineering, Cyber Security, and Artificial Intelligence & Data Science.

